

An Adaptive LQG Model of Human Learning in Surgical Robot Teleoperation

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Abstract—Surgical digital twins require models of the adapting human operator in addition to models of the robot and scene. Twenty-four participants practiced a ring-through-rail teleoperation task on the da Vinci Research Kit while robot kinematics were recorded. An adaptive linear-quadratic-Gaussian (LQG) controller performed matched rail-following tasks. The controller combines Kalman state estimation with optimal feedback control and maintains an internal estimate of the robot motion-scaling mapping. Prediction error updates this estimate over practice. Human trajectories became more direct with training, particularly in translational movement efficiency. The adaptive LQG controller showed a corresponding reduction in translational path length as its internal mapping estimate converged toward the true mapping. The comparison is qualitative rather than an individual-level fit, but both humans and the model exhibit the same principal signature of learning: progressively more efficient translational control with practice. Adaptive optimal control therefore provides an interpretable baseline for representing the changing operator within an operator-aware surgical twin, while remaining an idealized model rather than a complete account of surgical perception or expertise.

I. INTRODUCTION

Surgical teleoperation is a coupled human–robot control problem. The robotic plant maps console commands into instrument motion, while the operator observes the resulting motion and continuously corrects subsequent commands. Dynamic surgical-twin frameworks connect physical systems to computational models for monitoring, prediction, and assistance [1]. Yet even when the robot and task remain fixed, the coupled system changes during training because the operator becomes better calibrated to the interface. A surgical twin that represents only the robot and environment therefore omits a consequential state: what the operator has learned about the robot’s response to control input.

Optimal feedback control provides a compact framework for representing this changing operator. A linear-quadratic-Gaussian (LQG) controller separates state estimation under sensory and process uncertainty from feedback control that balances tracking error against control effort [2], [3]. Motion scaling is itself a human-in-the-loop design variable in surgical teleoperation [4]. We add an adaptive internal estimate of this mapping from human input to robot motion. A mismatch between the estimated and true mappings creates prediction error; learning reduces that error and changes subsequent con-

trol, consistent with prediction-error accounts of sensorimotor adaptation [5].

This workshop paper asks: *Can a compact adaptive optimal-control model reproduce a key signature of human learning during surgical robot teleoperation?* We compare human and simulated trajectories over matched training on a da Vinci Research Kit (dVRK) ring-through-rail task. The human data provide the empirical target; the goal is qualitative, mechanistic correspondence rather than fitting each participant. In this candidate operator-state representation, X_k is the fast, moment-to-moment teleoperation state, whereas g_{est} is a slower latent state describing calibration to the human–robot mapping. Allowing both to evolve gives an operator-aware twin a way to represent robot configuration and learning-related changes in human–robot coupling together. Participant retention, transfer, workload, and other post-training effects are reported in the full study [6] but are outside the empirical scope here; we use the learning rule only to derive an additional relearning prediction.

II. HUMAN EXPERIMENT AND ADAPTIVE MODEL

A. Teleoperation learning paradigm

Twenty-four participants used stereo endoscopic feedback and the dVRK surgeon console [7] to guide a ring along a fixed 3-D rail while avoiding contact (Fig. 1). The task is a controlled abstraction of surgical teleoperation and is included in the Fundamentals of Robotic Surgery curriculum [8]. The Georgia Institute of Technology Institutional Review Board approved the study (IRB2025-410), and all participants provided informed consent.

Three rails varied in curvature, elevation, and number of turns. Level 1 was a planar circle; Level 2 had two turns and two elevation levels; Level 3 had three turns and three elevation levels. Participants were randomly assigned to one rail and one motion-scaling ratio, 1:0.2 or 1:0.6, yielding six training conditions ($n = 4$ per condition). Each participant completed 12 training trials, grouped into four blocks of three. The full protocol also included pre- and post-training tests; they are not analyzed here.

Robot and console signals were recorded as synchronized time series. The principal learning outcome in this paper is robot translational path length, the distance traveled by the end effector during a trial. A shorter path indicates that fewer or

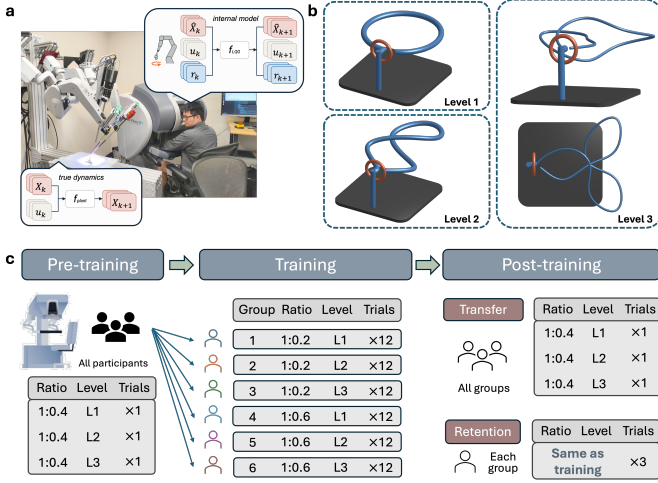


Fig. 1. Surgical teleoperation task and adaptive operator framing. (a) Participants controlled a dVRK to move a ring along a 3-D rail. The physical robot constitutes the true plant, while the internal operator model predicts how control commands change robot state. (b) Three rail geometries provided increasing spatial complexity. (c) The complete experiment included pre-training, 12 training trials in one of six rail-by-scaling conditions, and post-training tests. The present workshop analysis focuses only on learning during the 12 training trials.

smaller translational corrections were required to complete the same rail-following task. Rotational path length is retained as a contrast because translation and orientation need not change in parallel during learning.

B. Teleoperation plant and internal state estimate

We model the operator as an adaptive LQG controller acting on a rail-centric, discrete-time plant. The state

$$X_k = [p_k^\top, \psi_k^\top, \dot{p}_k^\top, \omega_k^\top]^\top \in \mathbb{R}^{12} \quad (1)$$

contains 3-D position p_k , orientation ψ_k , and their translational and angular velocities $\dot{p}_k$ and ω_k . The control input $u_k = [a_{t,k}^\top, a_{r,k}^\top]^\top \in \mathbb{R}^6$ contains translational and rotational accelerations. The true robot evolves as

$$X_{k+1} = AX_k + B(g_{\text{true}}u_k) + w_k, \quad (2)$$

$$y_k = CX_k + v_k, \quad (3)$$

where g_{true} is the motion-scaling gain implemented by the robot. Process disturbance $w_k \sim \mathcal{N}(0, W)$ and measurement noise $v_k \sim \mathcal{N}(0, V)$ are independent zero-mean Gaussian variables.

The controller does not initially act from the true mapping. Instead, its internal design model contains an estimate $g_{\text{est},k}$, which determines how strongly it expects a command to move the robot. A Kalman filter combines the model prediction with the observed state:

$$\hat{X}_{k+1}^- = A\hat{X}_k + B(g_{\text{est},k}u_k), \quad (4)$$

$$\hat{X}_{k+1} = \hat{X}_{k+1}^- + L(y_{k+1} - C\hat{X}_{k+1}^-), \quad (5)$$

where L is the steady-state Kalman gain. This observer makes uncertainty explicit: internal prediction and noisy measurement jointly determine the state estimate used for control.

C. Optimal feedback control

At each time step, the operator tracks a rail-aligned reference state r_k . With estimated tracking error $e_k = \hat{X}_k - r_k$, the controller selects

$$u_k = -Ke_k = -K(\hat{X}_k - r_k), \quad (6)$$

where the LQR feedback gain K minimizes the infinite-horizon objective

$$J = \mathbb{E} \left[\sum_{k=0}^{\infty} e_k^\top Q e_k + u_k^\top R u_k \right]. \quad (7)$$

The positive semidefinite matrix Q penalizes deviation from the rail reference; the positive definite matrix R penalizes control effort. Their balance determines how aggressively the model corrects error. Unlike a purely geometric rail follower, the LQG model links movement to an uncertain state estimate and an explicit cost on corrective action.

D. Prediction-error learning rule

Learning changes the internal motion-scaling estimate rather than fitting a separate trajectory at each trial. Participants were familiarized at motion scaling 0.4; training then changed the true mapping to 0.2 or 0.6. The operator representation is correspondingly calibrated at $g_{\text{est},0} = 0.4$ before the training change. Over one time step Δt , the translational velocity prediction error is approximately

$$\epsilon_k \approx (g_{\text{true}} - g_{\text{est},k})u_k\Delta t. \quad (8)$$

Gradient descent on squared prediction error gives the update

$$g_{\text{est},k+1} = g_{\text{est},k} + \eta(g_{\text{true}} - g_{\text{est},k})u_k^2\Delta t^2, \quad (9)$$

where $\eta > 0$ is the learning rate and u_k^2 denotes squared command magnitude in the adapted translational channel. Prediction errors are accumulated over a trial and the mean gradient updates g_{est} for the next trial. During training, the estimate therefore moves from 0.4 toward the practiced mapping, 0.2 or 0.6. As g_{est} approaches g_{true} , predicted command consequences become more accurate, reducing unnecessary corrective motion. Because the update is weighted by u_k^2 , equal-magnitude mapping errors need not yield equal adaptation rates: g_{est} changes the predicted robot response, which changes u_k , prediction error, and the subsequent update. This closed-loop consequence motivates the separate relearning prediction discussed below.

The simulated controller completed the same 12-trial training progression on the same three rail geometries and two scaling conditions as the human participants and received the rail centerline as its reference. We extracted translational and rotational path lengths from each simulated trajectory using the same conceptual measures as in the human analysis. Model parameters and sensitivity analyses are provided in the full report and accompanying code [6].

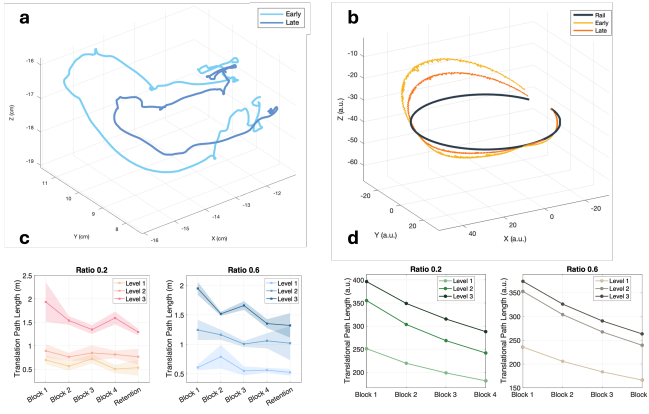


Fig. 2. Matched human and model learning dynamics. (a) Representative Level-1 human trajectory early and late in training; the late trajectory is more direct. (b) Early and late Level-1 trajectories generated by the adaptive LQG controller relative to the rail centerline; simulated motion becomes more efficient as the internal scaling estimate converges. (c) Human translational path length by training block and condition. The source graphic also displays retention at the far right; the present analysis concerns Blocks 1–4. (d) Simulated translational path length across the four matched training blocks. Humans and the model show qualitatively similar reductions in translational path length, not quantitative identity. Human shading denotes $\pm$ SEM.

III. MATCHED HUMAN–MODEL LEARNING DYNAMICS

A. Human trajectories become more efficient

Human teleoperation became progressively more efficient with practice (Fig. 2a,c). In the representative Level-1 example, the late trajectory contains fewer large deviations and follows a more direct path than the early trajectory. At the group level, translational path length generally decreased over the four training blocks across rail and scaling conditions. The full mixed-effects analysis, which included Blocks 1–4 and the subsequent retention observation, confirmed a block effect on translational path length, $F(4, 115) = 5.162$, $p = 0.001$ [6]. This statistic is reported only as evidence for the overall learning pattern; retention itself is not a result of the present workshop analysis.

The improvement was selective rather than a uniform contraction of every movement dimension. Rotational path length did not show a reliable block effect in the full analysis, $F(4, 115) = 0.706$, $p = 0.590$. The present adaptive state concerns translational motion scaling, so its strongest predicted effect is on translational correction efficiency. Wrist orientation may remain governed by rail geometry and task-specific orientation requirements even as translational calibration improves; we do not claim that the simulation quantitatively predicts this rotational null.

B. Adaptive LQG reproduces the qualitative learning signature

The adaptive LQG controller generated the corresponding model signature (Fig. 2b,d). Early in training, an inaccurate g_{est} caused systematic disagreement between predicted and realized robot motion. Prediction-error updates brought the estimate toward g_{true} across trials. As calibration improved,

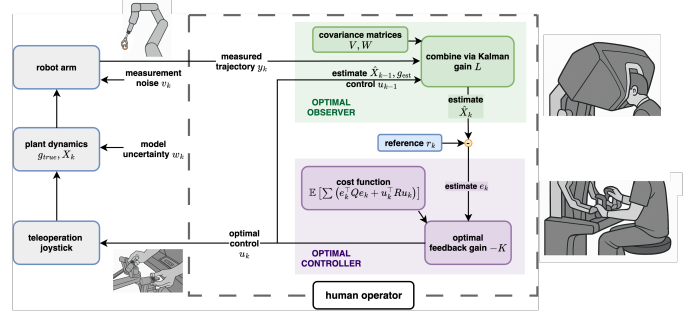


Fig. 3. Adaptive optimal-control representation of the human operator. The LQG operator combines Kalman-filter state estimation with optimal feedback control under measurement and process uncertainty. Its internal mapping estimate g_{est} determines predicted command consequences and is updated from prediction error.

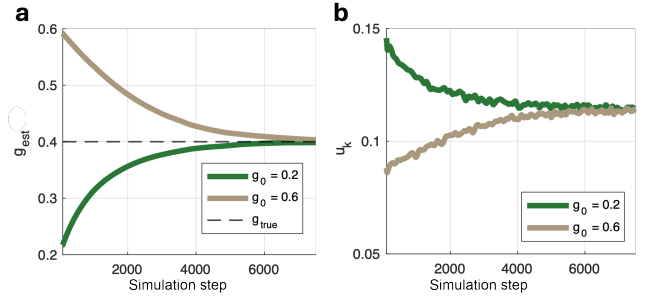


Fig. 4. Model-derived prediction for relearning after a subsequent motion-scaling change. After training calibrates g_{est} toward 0.2 or 0.6, g_{true} changes to 0.4. (a) The estimates converge asymmetrically. (b) Underestimation elicits larger commands u_k than overestimation; because the update in Eq. (9) is weighted by u_k^2 , larger commands produce faster relearning. Simulation curves are mechanistic illustrations, not participant fits.

the same optimal-feedback architecture produced shorter, more direct translational trajectories and required fewer large corrections.

This matched pattern appears at both scales shown in Fig. 2. At the trajectory level, later human and simulated paths are more direct than their early counterparts. Across training blocks, translational path length decreases for both humans and the model. The absolute units and trajectory geometry differ, so the curves should not be interpreted as a quantitative identity. The simulation was not fit to each participant’s learning curve, and the model has privileged access to the rail centerline. The result is instead a qualitative and mechanistic correspondence: adaptation of one interpretable mapping state reproduces a principal signature of human teleoperation learning.

IV. DISCUSSION

The training comparison supports a narrow claim with direct relevance to operator-aware surgical twins: operator adaptation can be represented as a changing computational state. Figure 3 shows the closed-loop architecture. Here, X_k is the fast robot/task state, whereas g_{est} is the slower operator-calibration state. A robot/scene twin describes what the system is doing

now; g_{est} compactly represents how human–robot coupling has changed through learning. Unlike a retrospective performance score, this state participates in the generative control loop by changing state prediction and feedback commands.

Figure 4 adds a model-derived, testable prediction rather than a participant-level transfer analysis. After training calibrates g_{est} toward 0.2 or 0.6, a subsequent change in g_{true} to 0.4 creates equal-sized upward and downward mapping errors. When $g_{\text{est}} < g_{\text{true}}$, however, the controller expects weaker robot motion and issues larger commands than when $g_{\text{est}} > g_{\text{true}}$ (Fig. 4b). The u_k^2 weighting in Eq. (9) then produces a larger update and faster relearning from the underestimated mapping (Fig. 4a). The asymmetry thus emerges from the closed loop rather than being imposed manually. It provides a falsifiable prediction for future human experiments: equivalent upward and downward mapping errors should yield systematically different adaptation rates when corrective command magnitudes differ.

The model is intentionally limited. The sample comprised 24 university students, with four participants per rail-by-scaling condition, rather than surgeons; the ring-through-rail task is an abstraction rather than a clinical procedure. Humans inferred task state from stereo endoscopic images, whereas the controller received the rail centerline directly. It does not represent visual inference, surgical expertise, bimanual coordination, force or deformable-tissue interaction, fatigue, or strategic exploration. Nor was it fit to individuals: human and simulated trajectories differ in absolute scale and geometry, and Fig. 2 establishes only correspondence in the direction and progression of translational improvement. The model is therefore a control-theoretic baseline for one human component of a surgical twin, not a complete twin of a surgeon or operation.

These limitations define a validation path. An operator-aware twin could identify g_{est} from each person’s command and robot-motion history, combine it with uncertain scene estimates rather than a known reference, and test prospectively whether the inferred state predicts subsequent error or performance. Such participant-specific system identification is needed before the model could support personalized practice or interface settings.

V. CONCLUSION

Matched human and adaptive-LQG simulations show that a compact control-theoretic operator model can reproduce an important qualitative signature of surgical teleoperation learning: progressively more efficient translational control with practice. Its latent calibration state provides an interpretable candidate representation of the changing operator within a surgical digital twin, while the same mechanism generates a testable prediction about asymmetric relearning. The model is not a complete account of surgical skill or perception.

DATA AND CODE

De-identified data and MATLAB analysis code are available at <https://osf.io/acuyr/>.

REFERENCES

- [1] L. Asciak *et al.*, “Digital twin assisted surgery, concept, opportunities, and challenges,” *npj Digital Medicine*, vol. 8, Art. no. 32, 2025, doi: 10.1038/s41746-024-01413-0.
- [2] E. Todorov, “Stochastic optimal control and estimation methods adapted to the noise characteristics of the sensorimotor system,” *Neural Comput.*, vol. 17, pp. 1084–1108, 2005.
- [3] E. Todorov and M. I. Jordan, “Optimal feedback control as a theory of motor coordination,” *Nat. Neurosci.*, vol. 5, pp. 1226–1235, 2002.
- [4] S.-K. Hsia, Y.-H. Chuang, and C.-W. Chen, “Auto-tuned motion scaling in teleoperation based on human reaction model identification,” *IEEE Robot. Autom. Lett.*, vol. 7, no. 1, pp. 318–325, 2022, doi: 10.1109/LRA.2021.3126342.
- [5] Y.-W. Tseng, J. Diedrichsen, J. W. Krakauer, R. Shadmehr, and A. J. Bastian, “Sensory prediction errors drive cerebellum-dependent adaptation of reaching,” *J. Neurophysiol.*, vol. 98, no. 1, pp. 54–62, 2007, doi: 10.1152/jn.00266.2007.
- [6] Y. Huang, Y. Cai, M. Li, Y. Chen, and R. C. Wilson, “Human learning dynamics during teleoperation of surgical robots,” *npj Science of Learning*, published online Sep. 2, 2026, doi: 10.1038/s41539-026-00442-6.
- [7] P. Kazanzides *et al.*, “An open-source research kit for the da Vinci surgical system,” in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 6434–6439.
- [8] R. M. Satava *et al.*, “Proving the effectiveness of the Fundamentals of Robotic Surgery skills curriculum: A single-blinded, multispecialty, multi-institutional randomized control trial,” *Ann. Surg.*, vol. 272, pp. 384–392, 2020.